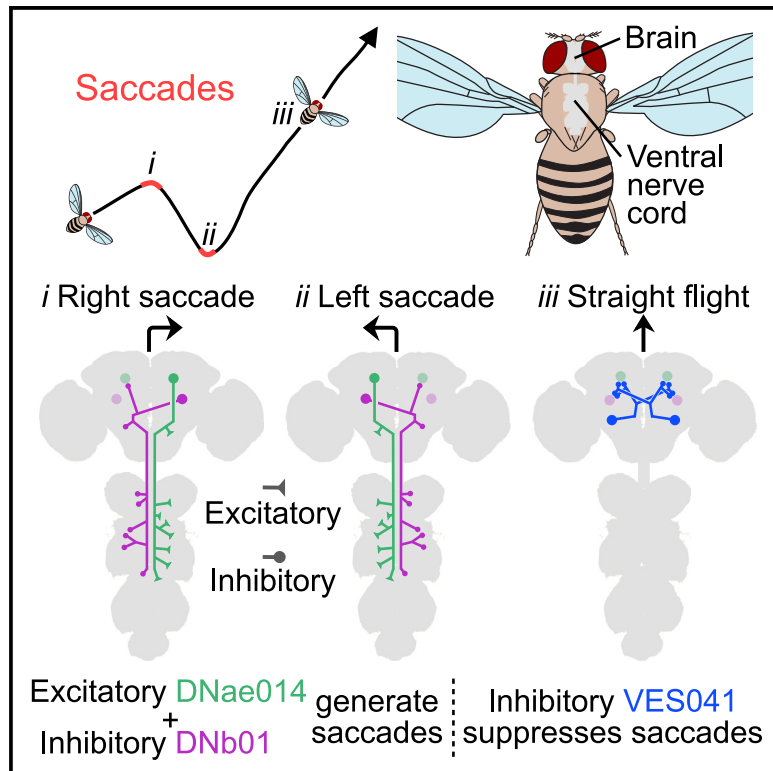


Current Biology

Descending control and regulation of spontaneous flight turns in *Drosophila*

Graphical abstract



Authors

Ivo G. Ros, Jaison J. Omoto,
Michael H. Dickinson

Correspondence

flyman@caltech.edu

In brief

The generation and regulation of flight turns play a critical role in foraging ecology. Ros et al. functionally characterize the neural network that generates spontaneous turns in fruit flies. A specialized neuron inhibits this network and suppresses flight turns, suggesting it may regulate the transition between local search and dispersal.

Highlights

- The activity of two pairs of descending neurons drives flight turns called saccades
- Couplets of one excitatory and one inhibitory descending cell form functional units
- An array of excitatory and inhibitory neurons provides input to the saccade network
- VES041 inhibits saccade network, possibly regulating state transitions during flight

Article

Descending control and regulation of spontaneous flight turns in *Drosophila*

Ivo G. Ros,¹ Jaison J. Omoto,¹ and Michael H. Dickinson^{1,2,*}

¹Division of Biology and Bioengineering, California Institute of Technology, 1200 E. California Blvd., Pasadena, CA 91125, USA

²Lead contact

*Correspondence: flyman@caltech.edu

<https://doi.org/10.1016/j.cub.2023.12.047>

SUMMARY

The clumped distribution of resources in the world has influenced the pattern of foraging behavior since the origins of locomotion, selecting for a common search motif in which straight movements through resource-poor regions alternate with zig-zag exploration in resource-rich domains. For example, during local search, flying flies spontaneously execute rapid flight turns, called body saccades, but suppress these maneuvers during long-distance dispersal or when surging upstream toward an attractive odor. Here, we describe the key cellular components of a neural network in flies that generate spontaneous turns as well as a specialized pair of neurons that inhibits the network and suppresses turning. Using 2-photon imaging, optogenetic activation, and genetic ablation, we show that only four descending neurons appear sufficient to generate the descending commands to execute flight saccades. The network is organized into two functional units—one for right turns and one for left—with each unit consisting of an excitatory (DNae014) and an inhibitory (DNb01) neuron that project to the flight motor neuropil within the ventral nerve cord. Using resources from recently published connectomes of the fly, we identified a pair of large, distinct interneurons (VES041) that form inhibitory connections to all four saccade command neurons and created specific genetic driver lines for this cell. As predicted by its connectivity, activation of VES041 strongly suppresses saccades, suggesting that it promotes straight flight to regulate the transition between local search and long-distance dispersal. These results thus identify the key elements of a network that may play a crucial role in foraging ecology.

INTRODUCTION

When searching their environment, many species of flies execute a series of straight flight segments interspersed with extremely rapid turns lasting less than ~50 ms. When first identified, these brief turns were termed saccades, in analogy with the rapid eye movements of primates.¹ In flies, however, the entire body rotates during each saccade, not just the eyes. Saccades are thought to benefit flying animals in several ways. From a sensory perspective, saccades may restrict the deleterious effects of motion blur to brief moments interjected within longer sequences of gaze stabilization.^{2–4} Brief bursts of saccades in the same direction may aid local search strategy by allowing the animal to quickly scan the local environment for salient visual and olfactory features.⁵ More recently, it has been suggested that comparing sensory measurements immediately before and after each saccade might enable flies to estimate key parameters that are otherwise not directly measurable, such as the direction and magnitude of the ambient wind.^{6,7} For all these hypotheses, the timing between saccades is critical, and several factors are thought to influence the frequency of the spontaneous turns. For example, tethered flies flying in a closed-loop configuration will greatly reduce their saccade rate and instead actively steer toward salient visual objects such as vertical edges—a robust fixation behavior that they maintain for several hours.⁸ Similarly, flies also reduce saccade rates in the presence of a small bright spot,

which they presumably interpret as the sun. In this case, however, they do not necessarily steer directly toward the distant bright object but rather maintain it at an arbitrary azimuthal angle.⁹ Patterns of polarized light that mimic those in the natural sky induce a similar reflex.^{10,11} This orientation behavior using celestial cues, termed *menotaxis*, allows flies and other insects¹² to maintain a constant heading over long distances (<10 km) while dispersing.^{13,14} Flies also suppress saccades in the presence of a constant odor stream to promote their progress upwind toward a food source.^{15–17} Despite its importance with respect to multiscale search, the neural mechanisms that underlie both the generation of saccades and their suppression during dispersal and plume tracking remain largely unknown.

Within the central nervous system of insects, descending neurons (DNs) constitute a critical stage in the transformation of sensory input in the brain into motor commands in the ventral nerve cord (VNC).¹⁸ *Drosophila* possess ~650 pairs of DNs,^{19–21} some of which appear to function as specialized command neurons for specific behaviors, including courtship,^{22,23} walking backward,²⁴ turning,²⁵ take-off,^{26,27} and landing.²⁸ Thus, DNs provide a logical starting point for investigating the circuits that generate and regulate flight saccades. A prior study using electrophysiology and 2-photon imaging implicated a pair of DNs, called AX, in the execution of spontaneous and visually elicited flight saccades.²⁹ However, this work employed a broad GAL4-driver line (R65G08), and a separate study found no significant effect on saccade rate when

the cells within this line were silenced via a temperature-sensitive dynamin mutation that blocks synaptic transmission.³⁰ To confirm and further investigate the role of AX in saccade generation, we constructed a sparse split-GAL4 driver for this neuron that was better suited for functional imaging, optogenetic activation, and genetic ablation. From here on, we will refer to the AX neuron as DNae014, as this is its identifier in the FlyWire database.^{31,32} Because we suspected that multiple pairs of DNs might be involved in driving the bilateral wing movements exhibited during saccades, we screened a large collection of split-GAL4 lines²¹ for additional cells that function to trigger rapid turns. Our screen led to the discovery of one additional cell (DNb01) that, together with DNae014, appears to form two functional units that drive saccades to the left and right. We systematically analyzed the upstream inputs to these DNs using emerging connectomes of the *Drosophila* brain^{31–34} and found a bilateral pair of large inhibitory interneurons (VES041) whose activation suppresses saccades, thereby promoting straight flight. Thus, we have identified the core cellular components of a network that generates and regulates spontaneous flight maneuvers in flying *Drosophila*.

RESULTS

To confirm the role of DNae014 in generating flight saccades, we created a split-GAL4 line³⁵ that targets this neuron for functional imaging, optogenetic activation and silencing, and genetic ablation. The split-GAL4 line we created was quite sparse (Video S1), although we occasionally saw faint expression of other cells, including one unidentified DN. To characterize the activity of individual DNae014 neurons, we used 2-photon functional imaging to record from the arbors of DNae014 cells on the left and right sides of the brain, while simultaneously tracking wing motion during flight (Figures 1A–1E). The difference in GCaMP7f fluorescence between the bilateral pair of cells (R-L $\Delta F/F$) correlated remarkably closely with the difference in wing stroke amplitude (L-R WSA) (Figures 1F and 1H). We used a simple classifier to determine the timing of saccades based on transient differences in WSA (dots in Figures 1H and 1I). We interpreted these saccades as spontaneous events because our experiments were conducted in darkness. We also recorded from the arbor of the weakly expressing, off-target DN mentioned above but did not detect any functional activity during flight.

In addition to DNae014, we identified another cell, DNb01, whose activity is also tightly correlated with the production of spontaneous saccades (Figures 1D, 1E, 1G, and 1I). The split-GAL4 driver line for this cell (SS02383)²¹ is also very sparse, although the line does target some neurons that arborize in the ocellar ganglion (Video S2). Across individuals, the bilateral difference in cell fluorescence (R-L $\Delta F/F$) explained $89\% \pm 4\%$ (for DNae014) and $90\% \pm 3\%$ (for DNb01) of the variance in differential wing stroke amplitude (L-R) (Figure 1K). The activity levels between the left and right DNae014 cells followed an inverse, highly non-linear relationship, analogous to “flip-flop” components in digital electronic circuits (Figure 1J) and reminiscent of neurons identified in the steering behavior of male silkmoths.³⁸ The relationship for the two DNb01 cells was also inverse, but much more linear. Given the complex relationship between the electrical activity of a neuron and its GCaMP signal, these subtle differences in the bilateral patterns of the DNae014

and DNb01 cells may not be functionally relevant; nevertheless, the activities of both neurons correlate remarkably well with ipsilaterally directed saccades (Figure 1I).

Despite similar activity patterns, DNae014 and DNb01 are morphologically distinct. DNae014 possesses smooth, dendritic processes in the lateral accessory lobe (LAL) and superior posterior slope (SPS), with varicose terminals in the gnathal ganglion (GNG) and wing neuropil of the VNC (Figures 1C and 1D). As described before, based on dye fills following whole-cell recordings,²⁹ all processes of the DNae014 cell are ipsilateral to the soma in both the brain and VNC. In contrast, DNb01 crosses the midline via the LAL commissure with varicose terminals in the contralateral SPS, posterior ventrolateral protocerebrum, and LAL and sends a descending axon to the wing neuropil of the VNC contralateral to the cell body (Figures 1D and 1E). Both cells arborize extensively within the LAL, a region implicated in turning behaviors in a variety of insect species.^{39,40} Although we did not record from DNae014 and DNb01 simultaneously, transient changes in the activity of either corresponds with a rapid alteration in wing motion and, conversely, every change in wing motion corresponds with transients in the activity of the neurons. Based on this close correlation (Figure 1K), we tentatively conclude that these two pairs of DNs are co-active during flight saccades (Figure 1).

To confirm the prior observation that DNae014 responds to expanding visual patterns and test whether DNb01 exhibits a similar phenomenology, we imaged bilateral activity of the DNs while presenting a looming stimulus at different azimuthal positions (Figure 2). For both DNae014 and DNb01, a looming spot presented in one visual hemisphere increased activity in cells on the opposite side (Figure 2B). Although the absolute magnitude of the measured $\Delta F/F$ response for DNae014 was larger, this result based on Ca^{2+} imaging need not reflect an actual difference in the membrane-level responses of the two DNs. Recently, other DNs have been identified that appear to mediate looming responses during flight,⁴¹ which suggests that DNs other than DNae014 and DNb01 may be involved in visually elicited avoidance reactions.

To test whether DNae014 and DNb01 are sufficient to elicit saccades, we used several methods to activate individual neurons on either the left or right side of the brain while measuring the resulting changes in wing motion. First, we drove CsChrimson bilaterally in DNae014 cells but used focal 2-photon excitation to unilaterally activate neurons on one side of the brain or the other (Figures 3A–3C). Activation of the left DNae014 cell reliably elicited a saccade to the left and activation of the right cell elicited a saccade to the right, consistent with the correlation recorded during spontaneous activity (Figure 1). These evoked syndirectional saccades also confirm prior experiments that used focal injection of ATP to activate ectopically expressed P_2X_2 channels⁴² in DNae014.²⁹ Focal 2-photon CsChrimson activation was not feasible for single DNb01 cells, however, because each neuron branches extensively in both the left and right hemispheres (Figures 1D and 1E). Instead, we unilaterally activated the DNs using the sparse predictive activity through recombination competition (SPARC) system⁴³ to express CsChrimson (Figure 3D). Because the SPARC reagent produced flies with CsChrimson in either both targeted cells, neither cell, the left cell only, or the right cell only, we scored the expression pattern of each individual after testing their behavioral responses to pulses

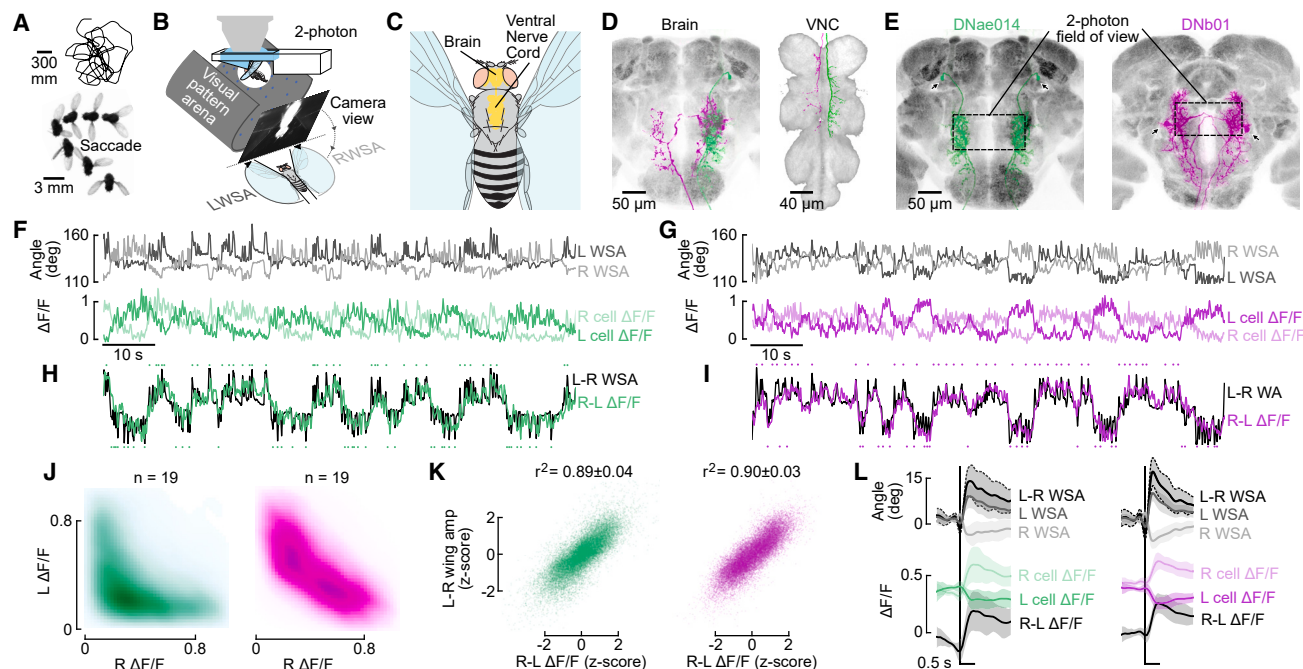


Figure 1. Activities of DNae014 and DNb01 correlate strongly with spontaneous saccades

(A) Flight trajectory and photomontage illustrating a single saccade, adapted from Censi et al.³⁶ and Muijres et al.³⁷
 (B) Schematic of set-up for monitoring neural activity with GCaMP7f and simultaneously tracking wing stroke amplitudes during flight.
 (C) Fly with central nervous system.
 (D) Anatomy of right DNae014 (green) and DNb01 (magenta) cells in the brain and VNC with nc82 staining in gray.
 (E) Bilateral expression patterns of DNae014 and DNb01 cells in the brain from split-GAL4 lines, showing the fields of view scanned in two-photon experiments (dashed boxes). Black arrows indicate somata.
 (F) Top: left (dark gray) and right (light gray) wing stroke amplitudes. Bottom: normalized GCaMP7f fluorescence ($\Delta F/F$) in left and right DNae014 cells during a 90-s flight epoch.
 (G) Similar to (F) but for DNb01.
 (H) Right-left difference in DNae014 signals (green) superimposed with left-right difference in wing stroke amplitudes (black). Both traces are normalized by Z score from the traces above in (F). Automatically detected right and left saccades depicted as green dots above and below, respectively.
 (I) Similar to (H), but for DNb01.
 (J) Pooled regressions for DNae014 (green) and DNb01 (magenta) of left versus right cell activity ($\Delta F/F$), normalized by Z score ($n = 19$ flies) and plotted as a kernel density estimate.
 (K) Regressions of L-R wing stroke amplitude versus R-L $\Delta F/F$, across 19 flies for DNae014 (green; $r^2 = 0.89 \pm 0.04$, mean $\pm$ SD) and DNb01 (magenta; $r^2 = 0.90 \pm 0.03$, mean $\pm$ SD).
 (L) Changes in DNae014 (left column) and DNb01 (right column) cell activity aligned to the onset of spontaneous flight saccades (vertical line). Presented as right saccades. Top traces: baseline-subtracted wing stroke amplitudes of left (dark gray), right (light gray) wings, along with left-right difference (black). Bottom traces: $\Delta F/F$ signals for right cell (light color), left cell (dark color), and right-left difference (black). Solid lines indicate mean of means for all individuals ($n = 19$ flies for each dataset); shaded areas indicate bootstrapped 95% CIs.
 See also [Figure S1](#) and [Videos S1](#) and [S2](#).

of 617 nm light. A distinct advantage of the SPARC method is that it enforces a blinded experimental design, with the subset of flies in which neither of the targeted cells expressed CsChrimson serving as a convenient control group. Unilateral activation of DNae014 evoked the expected changes in wing motion based on our 2-photon experiments ([Figure 3G](#)), and unilateral activation of DNb01 also evoked syndirectional saccades. However, the saccades evoked by DNb01 activation were smaller in amplitude and longer in time course than those evoked by DNae014 activation, suggesting that the two cells drive different components of the saccade motor program within the VNC.

To assess the necessity of the two pairs of DNs to generate spontaneous saccades, we silenced the neurons via expression of the light-gated GtACR1 channel. We periodically exposed

rigidly tethered flies expressing GtACR1⁴⁴ in DNae014 to 530 nm light pulses ([Figures S2A](#) and [S2B](#)). Although the experimental group exhibited a strong decrease in saccade rate when exposed to 530 nm light, empty-vector control flies, which did not express GtACR1, exhibited a substantial increase in saccade rate in response to the pulses of green light ([Figures S2C](#) and [S2D](#)). Whereas these silencing results support the hypothesis that suppressing DNae014 activity reduces the frequency of spontaneous saccades, the interpretation is complicated by the large increase in saccade rate exhibited by control flies in response to the activation light. For this reason, we adopted an alternative strategy to test the necessity of the two DNs to saccade by genetically ablating the cells through ectopic expression of the pre-apoptotic genes *reaper* and *hid*.⁴⁵

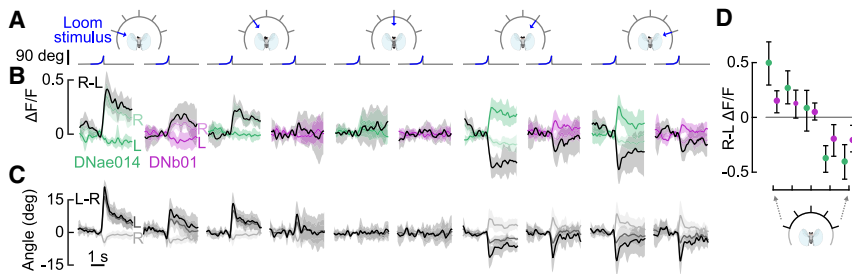


Figure 2. DNae014 and DNb01 are active during saccades elicited by visual loom

(A) Visual azimuth directions of looming stimuli from -72° , -36° , 0° , 36° , and 72° (left to right). Traces below indicate the time course of stimulus size. (B) Normalized GCaMP7f fluorescence for DNae014 (left panels, green, $n = 9$) and DNb01 (right panels, magenta, $n = 12$) in response to each stimulus direction for left cell (L), right cell (R), and bilateral differential (black, R-L). Solid lines represent the grand mean of the mean traces for individual flies, the shaded areas indicated the bootstrapped 95% confidence intervals (CIs).

(C) Wing stroke amplitude responses to every stimulus direction of left (L), right (R), and differential (L-R) wing stroke amplitudes. Solid lines represent the grand mean of the mean traces for individual flies; the shaded areas indicate the bootstrapped 95% confidence intervals (CIs).

(D) Bilateral difference (R-L) $\Delta F/F$ for peak stimulus responses to each lobe direction. Error bars indicate standard deviations of individual means. DNae014 (green) responded more strongly but with a similar sign relative to stimulus direction, compared with DNb01 (magenta). Peak responses for DNae014 $\Delta F/F$ were larger than those for DNb01 ($p < 0.001$; two-way ANOVA with cell type and lobe direction as the main effects and individual as a random effect).

See also Videos S1 and S2.

We quantified flight saccades in rigidly tethered flies in which either DNae014 or DNb01 was ablated. Empty-vector control flies exhibited a saccade frequency of 0.48 ± 0.18 Hz (mean $\pm$ SD), consistent with prior studies under both free and tethered flight conditions.^{29,46,36} Genetic ablation of DNae014 reduced saccade frequency to 0.09 ± 0.09 Hz ($p < 0.001$, Kruskal-Wallis test followed by post hoc comparisons with Tukey's HSD method), whereas ablation of DNb01 reduced the rate to 0.12 ± 0.14 Hz ($p < 0.001$; Figures 4B and 4C). Thus, ablation of either DNae014 or DNb01 significantly reduced the rate of spontaneous saccades but did not eliminate them. This result suggests that bilateral pairs of either DNae014 or DNb01 acting alone can still generate partial saccades, albeit at a reduced spontaneous frequency, or alternatively, that spontaneous saccades that persist following ablation of either cell type might be generated by an alternate pathway.

To gain further insight into the effect of cell ablation on saccade generation, we repeated the experiments using a magnetotether arena, in which both rapid turns and epochs of straight flight are easier to observe and classify⁴⁶ (Figures 4D and 4E). Ablating the DNs reduced the saccade rate in the magnetotether from 0.37 ± 0.10 Hz in empty-vector controls to 0.20 ± 0.06 Hz for DNae014-ablated flies ($p < 0.001$) and 0.19 ± 0.10 Hz for DNb01-ablated flies ($p < 0.001$; Figures 4E and 4F). In addition, genetic ablation of either cell type resulted in a qualitatively different temporal pattern of flight behavior distinguished by long sequences of straight flight interspersed with rapid bursts of saccades, typically in the same direction for DNae014-ablated flies (Figures 4E, 4G, and 4I). We quantified this influence of cell ablation on saccade timing by determining the maximum intersaccade interval (ISI_{max}) for each 2-min trial of each fly. The ISI_{max} for control flies was 13.2 ± 4.9 s, compared with 33.1 ± 15.2 s for DNae014-ablated flies ($p < 0.0001$) and 33.1 ± 22.1 s for DNb01-ablated flies ($p < 0.0001$; Figure 4H). DNae014-ablated flies saccaded more frequently in the same direction as their preceding saccades, with a syndirectional saccade ratio (i.e., the proportion of saccades in one direction immediately following a saccade in that same direction) of 0.73 ± 0.13 compared with 0.55 ± 0.10 for control flies ($p = 0.0005$). The syndirectional saccade ratio for DNb01-ablated flies was 0.64 ± 0.20 , which was not significantly different from either controls or DNae014-

ablated flies (Figure 4I). As observed previously,⁴⁶ the speed and magnitude of saccades in the magnetotether were highly correlated, with faster saccades being larger (Figure 4J). The saccades after DNae014 ablation were slower than controls, as were those following DNb01 ablation ($p < 0.001$ and $p = 0.009$, respectively; Figure 4K). The altered saccade dynamics and temporal distribution after ablation support the hypothesis that each DN can produce saccades independently of the other but with different dynamics than those of control flies. The results support a working hypothesis that the two DNs play complementary roles by activating different components of the motor circuit in the VNC responsible for generating saccades. Together, functional imaging, unilateral activation, and ablation experiments suggest that two pairs of descending interneurons, DNae014 and DNb01, function together as saccade-generating units (SGUs) to execute commands for spontaneous turns during flight.

To further examine the underlying circuitry in the brain, we identified the two pairs of DNs in two available connectomes. Comparing confocal image stacks of GFP expression in DNae014 with the FlyWire and hemibrain connectomes^{31–34} revealed at least four DN pairs with similar morphology, all within the DNa class, as defined by Namiki and coworkers.²¹ Of these, only one (DNae014) appeared to be a precise match with the cell originally described as AX by Schnell and coworkers²⁹ (G. Jefferys, personal communication), which is why we chose to adopt the new nomenclature^{31,32} (K. Eichler, P. Brooks, T. Stürmer et al., personal communication). DNae014 appears to be identical to the PS017_R neuron in the hemibrain connectome. Our confocal microscopy images of DNb01 matched only one DN in the two brain connectomes, which was already annotated as DNb01 in both the hemibrain and FlyWire datasets. Based on the results of our physiological and behavioral experiments (Figures 1, 2, 3, and 4), we propose that the DNae014 and DNb01 cells with somata on the left side of the brain form a functional unit that generates leftward saccades, with the right DNae014 and DNb01 cells forming a unit that generates rightward saccades. Currently, it remains possible that additional DNs are involved in these SGUs. Neurotransmitter predictions based on machine learning⁴⁷ suggest that DNae014 is cholinergic and DNb01 is glutamatergic (Table S1), with the latter classification corroborated by *in situ* expression of VGLUT.¹⁹ These putative

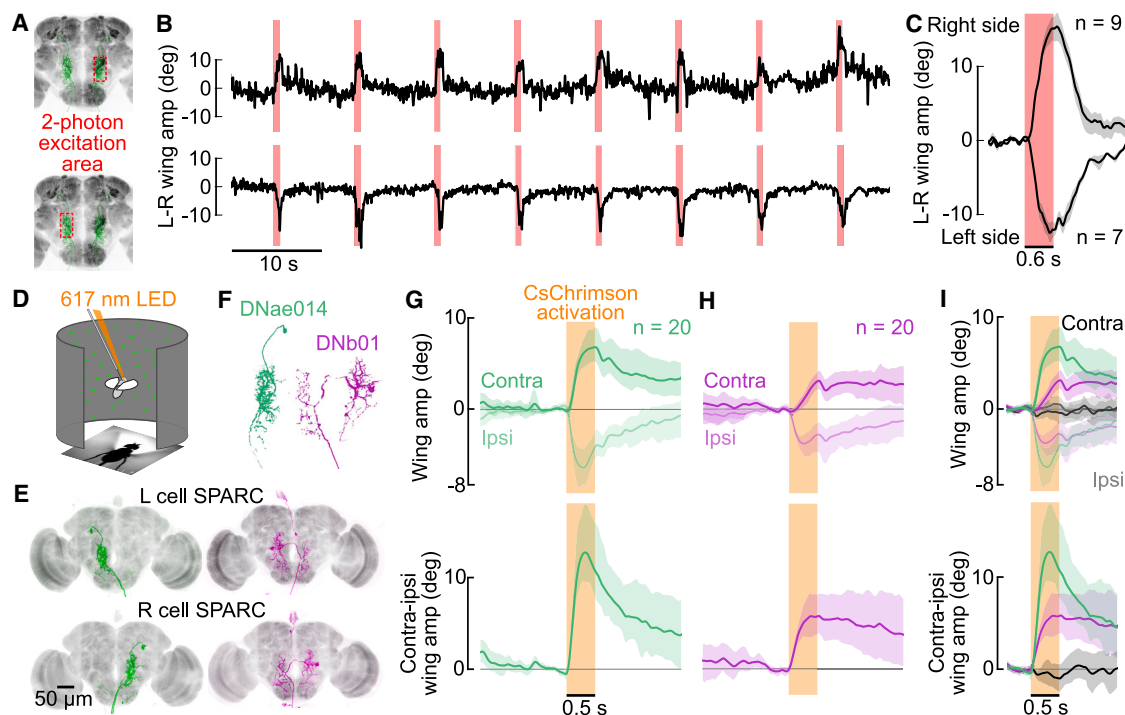


Figure 3. Unilateral activation of either DNae014 and DNb01 elicits directional saccades

(A–C) Unilateral 2-photon excitation of either left or right DNae014 neurons.

(A) Approximate 2-photon excitation areas of the right and left dendritic arbors of DNae014.

(B) Example traces of L–R wing stroke amplitude during unilateral CsChrimson excitation on the left (top) or right (bottom) side of brain.

(C) L–R wing stroke amplitudes aligned to the excitation light pulse. As in all panels, solid lines indicate the mean of all individual means, the shaded patches indicate the bootstrapped 95% CI; $n = 9$ flies for right stimulations, $n = 7$ flies for left stimulations.

(D–I) Unilateral activation of DNae014 or DNb01 neurons in rigidly tethered flies using split-GAL4 and SPARC expression of CsChrimson.

(D) Flight arena with 617 nm excitation light.

(E) Maximum intensity projection of TdTomato signal in examples of unilateral cell expression using SPARC with split-GAL4 drivers for DNae014 (green) and DNb01 (magenta). Nc82 staining shown in gray.

(F) Representative morphology of right DNae014 and DNb01 cells.

(G) Contralateral and ipsilateral wing stroke amplitudes (top) and contralateral-ipsilateral difference (bottom), aligned to activation pulse; $n = 20$ flies. Shaded patches indicate the bootstrapped 95% CI.

(H) Similar to (G), but for DNb01; $n = 20$ flies. Shaded patches indicate the bootstrapped 95% CI.

(I) Combined data from (G) and (H), including control flies (black traces, gray envelopes) in which neither cell expressed CsChrimson; $n = 20, 20$, and 19 for DNae014, DNb01, and controls, respectively. Shaded patches indicate the bootstrapped 95% CI.

See also Videos S1 and S2.

neurotransmitter types suggest that the two members of each unit collectively carry excitatory and inhibitory commands to opposite sides of the wing neuropil in the VNC. Recent analyses of the adult VNC in two new connectomes found that DNs connect to wing motor neurons both directly via monosynaptic inputs and indirectly via a large population of local premotor interneurons.^{19,48} Because both inhibitory and excitatory premotor neurons exist, the valence of a DN's neurotransmitter is not necessarily a clear predictor of its action on steering muscle activity during saccades.

Motivated by our hypothesis that each unilateral set of DNae014 and DNb01 cells functions as an SGU, we performed a connectivity search for neurons that provide direct inputs to both cells (Figure 5). Basing our analysis on the lowest synapse number threshold that maintained a bilaterally symmetrical connectivity pattern (see STAR Methods for details; Figure S3A), we identified eight neuron types that provide direct input to the

SGUs (a diverse population we call SGUs). Five of these SGUs are pairs of neurons with unique morphology and three constitute small sets of cell types with nearly identical morphology and connectivity (Figure S3B). In addition to making direct synapses with the two members of each SGU, several SGUs send a collateral that directly connects with other SGU members on the ipsilateral or contralateral side (Figure 5). We further identified a class of intermediary neurons that form strong connections between members of contralateral SGUs. Predictions of transmitter type based on machine learning suggest that the population of SGUs and intermediary cells includes both excitatory (acetylcholine) and inhibitory (glutamate or GABA) neurons (Figures 5 and S4). We also ranked the top 10 additional neuron types that form presynaptic connections with members of this putative saccade network. These inputs include neurons that likely carry sensory information from brain regions, including the lobula (Lo), posterior lateral protocerebrum (PLP),

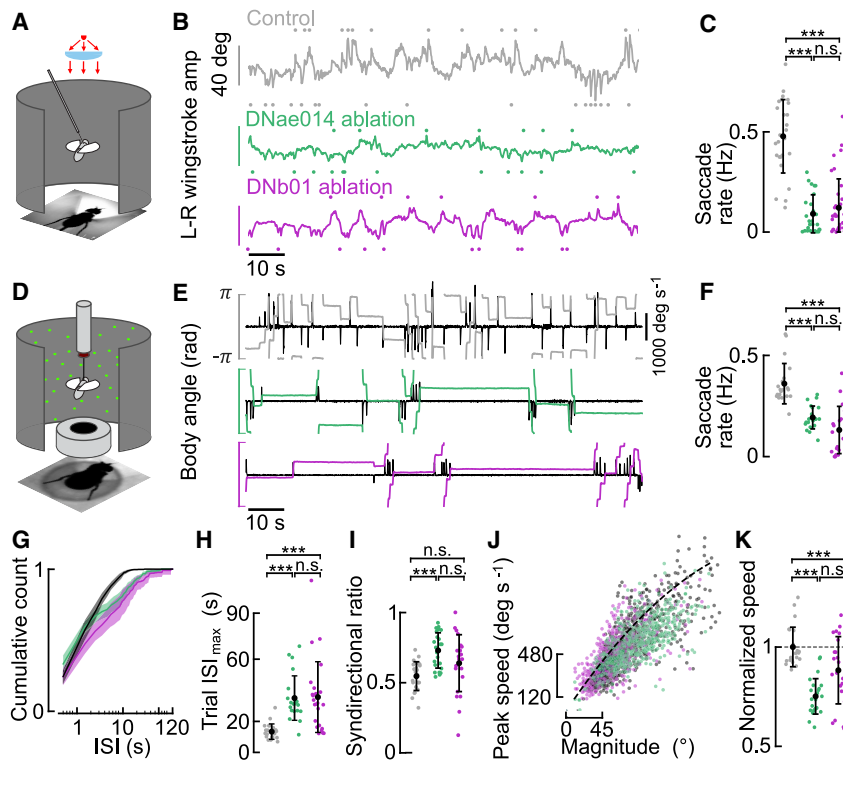


Figure 4. Ablation of either DNae014 or DNb01 alters saccade dynamics

(A) Rigid-tether flight arena with IR backlighting to track wing stroke amplitudes. (B) Transient changes in L-R wing stroke amplitude indicate fictive saccades. Example control fly (top, gray), DNae014-ablated fly (middle, green), and DNb01-ablated fly (bottom, magenta). (C) Individual saccade rates in the rigid tethered arena ($n = 26, 25$, and 36 , respectively). Error bars indicate standard deviations of individual means. (D) Magnetotether flight arena with IR backlighting to track body angle and a static visual pattern. (E) Example traces with saccades apparent as rapid changes in body angle and angular velocity transients (black traces). (F–K) Color codes as in (B). $n = 22, 23$, and 22 flies, respectively. (F) Saccade rates in the magnetotether arena. Error bars indicate standard deviations of individual means. (G) Cumulative histogram of inter-saccade intervals, plotted as mean of individual fly means. Shaded patches indicate the bootstrapped 95% CI. (H) Maximum inter-saccade interval (ISI_{max}) for each 2-min trial. Error bars indicate standard deviations of individual means. (I) Fraction of saccades syndirectional with the previous saccade. Error bars indicate standard deviations of individual means.

(J) Relationship between peak saccade speed and saccade magnitude. dashed line: Orthogonal-distance regression fit of a 2nd order polynomial to pooled data of control flies.

(K) Peak saccade speed normalized to fit in (J), plotted as mean of individual fly means. Error bars indicate standard deviations of individual means. Statistical differences were assessed using Kruskal-Wallis tests with Tukey's HSD method for post hoc comparisons (C, F, H, I, and K; *** $p < 0.001$; n.s., not significant).

See also [Figure S2](#) and [Videos S1](#) and [S2](#).

accessory optic tubercle (AOTu), and the LAL and central complex (CX) ([Figures 5I](#) and [S5](#)). A recent study also identified neuron types (PFL2 and PFL3) that carry information from the CX to DNs in the LAL.⁴⁹ In addition, DNb01 and several SGUs make recurrent inhibitory connections to network members on the contralateral and ipsilateral sides ([Figure 5A](#)). These anatomical pathways suggest that external sensory information and internal processing likely influence the dynamics of the network that generates spontaneous saccades within the LAL.

One cell, VES041, stood out in our analysis due to its unique feature of providing direct monosynaptic input to the DNae014 and DNb01 cells of both SGUs, as well as to several of their prominent upstream targets ([Figures 5C, 5E](#), and [S4](#)). The neurotransmitter for VES041 is predicted to be GABA, based on machine learning ([Table S1](#)). Thus, each of the paired VES041 cells could potentially inhibit all four cells that constitute the two SGUs and thereby strongly suppress the saccade rate. To test this, we generated two split-GAL4 lines that target VES041 neurons and used them to drive CsChrimson ([STAR Methods](#); [Figure 6A](#); [Video S3](#)). Presentation of 617 nm light strongly reduced spontaneous saccades in flies flying in the magnetotether apparatus from 0.52 ± 0.22 to 0.17 ± 0.10 Hz and from 0.71 ± 0.17 to 0.09 ± 0.05 Hz in the two driver lines we tested ([Figure 6](#)). In contrast, presentation of 617 nm light slightly increased the spontaneous saccade in control flies from 0.41 ± 0.16 to 0.68 ± 0.21 Hz

([Figure 6E](#)). The results are consistent with independent experiments conducted on rigidly tethered flies in which we classified saccades based on changes in wing stroke amplitude ([Figure S6](#)) during optogenetic activation of VES041. The simpler algorithm used for saccade classification from magnetotether data, based on the actual angular velocity of the fly's body, likely results in lower false-negative and false-positive rates compared with the method of classifying saccades performed by rigidly tethered flies. Nevertheless, despite the minor differences in the results from the activation experiments in the two different flight arenas, our findings were consistent across the two methods ([Figures 6](#) and [S6](#)) and thus support our hypothesis that VES041 is a potent inhibitor of spontaneous saccades. The VES041 neuron receives its strongest input from the vest (VES), GNG, LAL, SPS, flange (FLA), inferior bridge (IB), and saddle (SAD) ([Figures 6](#) and [S6F](#)), which suggests that this cell integrates both visual and olfactory information that is known to influence saccade rate.

DISCUSSION

We investigated the role of two pairs of DNs in the generation of body saccades in flying flies. Our results suggest a model in which two neurons, DNae014 and DNb01, form two SGUs. Although we could not record from DNae014 and DNb01

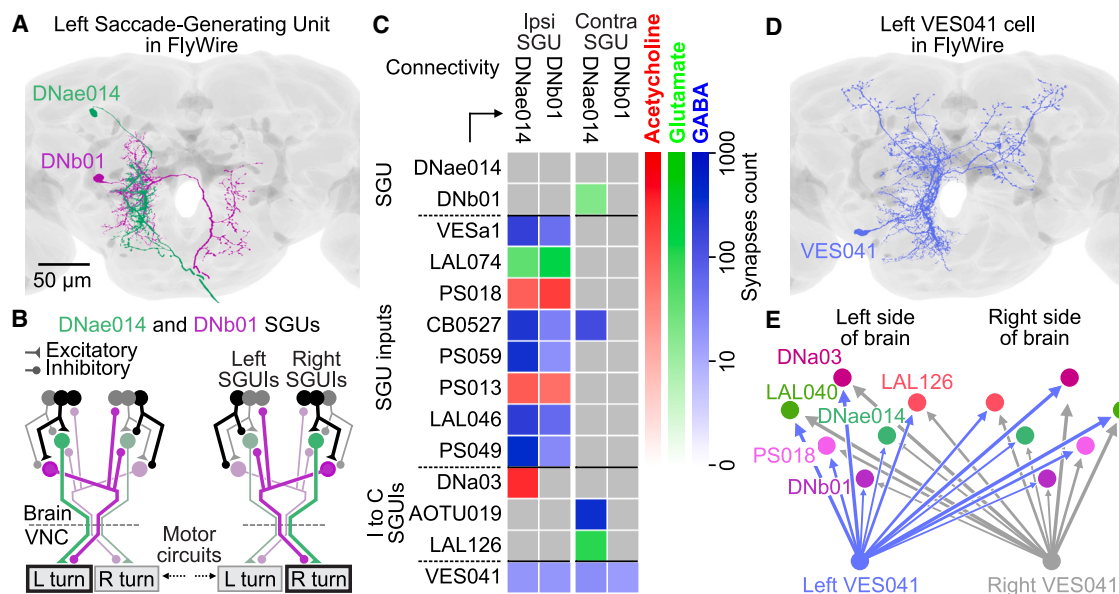


Figure 5. VES041 neurons innervate all DNae014 and DNb01 cells, along with several of their upstream targets

(A) Left saccade-generating unit (SGU) consisting of DNae014 (green) and DNb01 (magenta) cells (from FlyWire).

(B) Schematized model for network generating left and right saccades. Bright colors or black indicate active neurons; pale colors or gray indicate inactive neurons.

(C) Connectivity of putative members of the network that regulates directed saccades (rows) to SGU neurons DNae014 and DNb01 (columns; arrow indicates direction of information flow). Our analysis identified neurons that form input synapses to both SGU cells (SGUs), neurons that form connections between ipsilateral and contralateral SGUs (I to C SGUs), and a neuron that connects to all four SGU cells (VES041). Connectivity data are averaged assuming symmetrical arrangement of left and right network members.

(D) Morphology of the left VES041 cell in FlyWire.

(E) Connections of VES041 cells (blue for left; gray for right) to both SGUs and bilateral pairs of SGUs and intermediary neuron types. Colors match representations in Figures S4C–S4J. Line thickness is proportional to log of synapse count.

See also Figures S3–S5 and Tables S1–S3.

simultaneously, every rapid change in wing motion accompanied a transient change in the activity of either cell when recorded separately. This one-to-one correspondence supports the hypothesis that unilateral cells from DNae014 and DNb01 function together as single command units that carry both excitatory and inhibitory inputs to the premotor circuits in the VNC to execute flight saccades. In the brain, we used the DNae014 and DNb01 cells to identify a mosaic of excitatory and inhibitory neurons that innervate neuropils associated with steering and multi-modal sensory integration. One pair of large inhibitory interneurons, VES041, emerged as a likely candidate to suppress the execution of spontaneous turns because each cell makes strong inhibitory connections with all members of both the left and right SGUs. Optogenetic activation of VES041 strongly suppresses saccades, which suggests that it may play a regulatory role in natural behaviors, such as long-distance dispersal or upwind surges in response to an attractive odor. Although we have not yet succeeded in performing a viable double ablation experiment (i.e., silence or ablate both DNae014 and DNb01 in the same individuals), the results of VES041 activation serve as a rough proxy for simultaneous silencing of both DNae014 and DNb01 (Figures 6 and S6). Although the inhibition of saccades by activation of VES041 is parsimoniously explained by its GABAergic inputs to the DNae014 and DNb01 neurons, this tentative transmitter classification is based solely on machine learning and must be verified further using complementary techniques.

Although the high fidelity of the activity of both cells to the changes in wing motion is itself compelling (Figure 1), this phenomenon might result from strong ascending proprioceptive feedback or efferent copy. However, brief activation of either single DNae014 or single DNb01 cells reliably evoke syndirectional turns (Figure 3). Further, ablating either cell type results in unusually long sequences of straight flight interspersed by brief bursts of small amplitude saccades (Figure 4). The induction of long inter-saccade intervals provides intriguing evidence that these cells contribute directly to temporal patterning of the spontaneous saccades and do not simply relay upstream commands (Figures 1, 2, 3, and 4). However, ablating either cell alone does not abolish all spontaneous turns, even though the persisting saccades are slower. This partial loss-of-function effect might indicate that the motor commands resulting from either DNae014 or DNb01 activate sub-components of the complete motor program that are alone sufficient to generate partial, slower saccades (Figure 4J) or, alternatively, that the fly's nervous system contains as-yet unidentified parallel pathways for generating spontaneous turns. These alternative hypotheses are not mutually exclusive and, because our ability to cluster saccades according to type based on behavioral measurements is limited, not currently distinguishable.

Unilateral activation of both DNae014 and DNb01 evoked changes in wing motion similar to syndirectional saccades (Figure 3), consistent with the results from our functional imaging

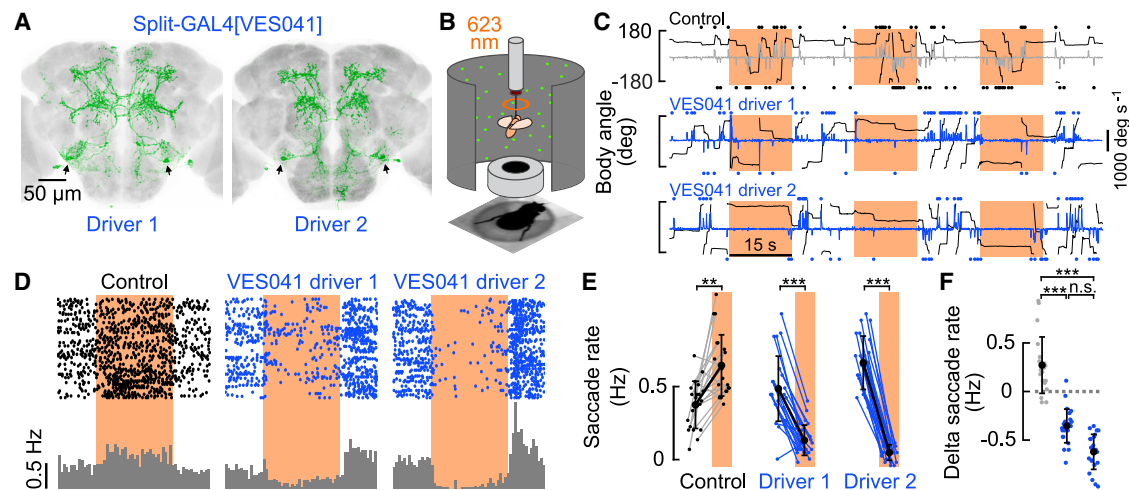


Figure 6. Activation of GABAergic VES041 neurons suppresses spontaneous saccades

(A) Split-GAL4 lines constructed to target VES041 (green, GFP); nc82 staining in gray.
(B) Optogenetic activation of VES041 in a magnotether arena using a 623-nm LED ring light (orange annulus in B and orange patches in C).
(C) Top: empty-vector control fly, with right and left saccades (dots above and below the panel) identified using angular velocity of body orientation (gray trace). Middle and bottom: activation of VES041 using drivers 1 and 2 (angular velocity in blue).
(D) Raster plots represent spontaneous saccades (black dots), during exposure to 623 nm light (orange patch). Each set of three rows are the presentations for an individual fly. Left: empty-vector split-GAL4 line driving CsChrimson; middle and right: VES041 split-GAL4 lines 1 and 2 driving CsChrimson. Gray histograms indicate mean saccade rate across populations.
(E) Individual saccade rates during light-off and light-on epochs for control and the two VES041 drivers; each line represents an individual fly ($n = 20, 24$, and 23 flies). Error bars indicate standard deviations of individual means.
(F) Individual differences in saccade rate from the light-off to light-on condition for the same groups as in (E). Error bars indicate standard deviations of individual means (** $p < 0.01$; *** $p < 0.001$; n.s., not significant; Kruskal-Wallis test followed by post hoc comparisons with Tukey's HSD method).
See also Figure S6, Table S1, and Video S3.

experiments (Figure 1). However, saccades evoked by DNb01 activation were smaller in amplitude and longer in time course than those evoked by DNae014 activation. These differences in magnitude of the motor response could be due to systematic differences in CsChrimson expression levels or cell-specific sensitivity to optogenetic activation and do not necessarily reflect inherent differences between the respective DNs. Nevertheless, when considered along with the difference in putative neurotransmitter types between DNae014 and DNb01, these nuanced differences in effects on wing motion suggest that the two cells in an SGU drive different components of the saccade motor program within the VNC.

The activity levels between the left and right DNae014 cells followed an inverse, highly non-linear relationship, analogous to “flip-flop” components in digital electronic circuits (Figure 1J) and reminiscent of neurons identified in the steering behavior of male silkmooths.³⁸ Further, the DNb01 cells synapse directly onto contralateral DNae014 cells. This is a simple reciprocal inhibitory motif, consistent with a network responsible for binary turning to either the left or the right (Table S2) and analogous with the arrangement of Mauthner cells in fish.⁵⁰ In addition, DNb01 and several SGUs make recurrent inhibitory connections to network members on the contralateral and ipsilateral sides (Table S2; Figure S5A). These anatomical pathways suggest that external sensory information and internal processing likely influence the dynamics of the network that generates spontaneous saccades within the LAL.

In summary, our results suggest that DNae014 and DNb01 form two SGUs that are inhibited by VES041. DNae014 and

DNb01 cells are probably members of a quite ancient network that is centered in the paired LALs of the brain and widespread among insects and other arthropods.^{40,51} In particular, comparative anatomical findings suggest these two pairs of DNs are homologous to flip-flop neurons responsible for the rapid turns exhibited during plume tracking behavior in the silkmooth, *Bombyx mori*.^{38,52} Future experiments are needed to identify the specific sensory cues and their anatomical pathways that influence VES041 activity and thus determine its specific role in regulating this ecologically important behavioral state transition in flying insects.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
 - Lead contact
 - Materials availability
 - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
- METHOD DETAILS
 - Anatomy
 - Functional imaging
 - Functional Imaging, regions of interest
 - Saccade detection

- Optogenetic activation experiments
- Unilateral genetic targeting of CsChrimson
- Connectomics
- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2023.12.047>.

ACKNOWLEDGMENTS

We thank members of our lab, including Ainul Huda, Will Dickson, Johan Melis, Francesca Ponce, and Matthew Clark. Additionally, we are grateful to Shigehiro Namiki, Sasha Rayshubskiy, Rachel Wilson, and Anna Ahn for helpful contributions and discussions and to Shigehiro Namiki, Wyatt Korff, and Gwyneth Card for sharing the driver lines used in the functional imaging screen. We thank Sasha Rayshubskiy for suggesting the presence of DNae014 in the GAL4 line VT025718. We thank the Drosophila Connectomics Group, including Greg Jefferis, Kathi Eichler, and Marta Costa, for sharing their large-scale proofreading and annotation, including all descending neurons, in FAFB-FlyWire prior to publication. Proofreading and annotation in Cambridge were supported by Wellcome Trust Collaborative Awards (203261/Z/16/Z and 220343/Z/20/Z) to G. Jefferis, NIH BRAIN Initiative grant 1RF1MH120679-01 to David Bock with G. Jefferis, and a Neuronex2 award to D. Bock and G. Jefferis (NSF 2014862, MRC MC_EX_MR/T046279/1). We specifically thank Laia Seratosa, Zeba Vohra, and Dharini Sapkal for proofreading and G. Jefferis and M. Costa for their help in identifying DNae014 neurons. We thank the Princeton FlyWire team and members of the Allen Institute for Brain Science for development and maintenance of FlyWire (supported by BRAIN Initiative grants MH117815 and NS126935 to Mala Murthy and Sebastian Seung). We also acknowledge members of the Princeton FlyWire team and the FlyWire consortium for neuron proofreading and annotation, specifically Jet Ivan Dolorosa from the Murthy and Seung labs. This research was supported by NIH grants U19NS104655 and 1R21NS106471-01A1 to M.H.D.

AUTHOR CONTRIBUTIONS

Conceptualization, I.G.R. and M.H.D.; methodology, I.G.R., J.J.O., and M.H.D.; software, I.G.R.; validation, I.G.R. and J.J.O.; formal analysis, I.G.R.; investigation, I.G.R. and J.J.O.; resources, I.G.R. and J.J.O.; data curation, I.G.R.; writing – original draft, I.G.R. and M.H.D.; writing – review & editing, I.G.R. and M.H.D.; visualization, I.G.R. and J.J.O.; funding acquisition, M.H.D.; supervision, M.H.D.; project administration, M.H.D.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: October 13, 2023

Revised: December 12, 2023

Accepted: December 14, 2023

Published: January 15, 2024

REFERENCES

1. Collett, T.S., and Land, M.F. (1975). Visual control of flight behaviour in the hoverfly, *Syrphia pipiens* L. *J. Comp. Physiol.* 99, 1–66.
2. Cellini, B., Salem, W., and Mongeau, J.M. (2021). Mechanisms of punctuated vision in fly flight. *Curr. Biol.* 31, 4009–4024.e3.
3. Land, M.F. (1999). Motion and vision: why animals move their eyes. *J. Comp. Physiol. A* 185, 341–352.
4. Schilstra, C., and van Hateren, J.H. (1998). Stabilizing gaze in flying blowflies. *Nature* 395, 654.
5. Heisenberg, M., and Wolf, R. (1984). Vision in *Drosophila*. In *Genetics of Microbehaviour* (Springer Verlag).
6. van Breugel, F., Jewell, R., and Houle, J. (2022). Active anemosensing hypothesis: how flying insects could estimate ambient wind direction through sensory integration and active movement. *J. R. Soc. Interface* 19, 20220258.
7. Stupski, S.D., and van Breugel, F. (2023). Wind gates search states in free flight. *bioRxiv*.
8. Götz, K.G. (1987). Course-control, metabolism and wing interference during ultralong tethered flight in *Drosophila melanogaster*. *J. Exp. Biol.* 128, 35–46.
9. Giraldo, Y.M., Leitch, K.J., Ros, I.G., Warren, T.L., Weir, P.T., and Dickinson, M.H. (2018). Sun navigation requires compass neurons in *Drosophila*. *Curr. Biol.* 28, 2845–2852.e4.
10. Warren, T.L., Weir, P.T., and Dickinson, M.H. (2018). Flying *Drosophila melanogaster* maintain arbitrary but stable headings relative to the angle of polarized light. *J. Exp. Biol.* 221, jeb177550.
11. Weir, P.T., and Dickinson, M.H. (2012). Flying *Drosophila* orient to sky polarization. *Curr. Biol.* 22, 21–27.
12. El Jundi, B., Foster, J.J., Khaldy, L., Byrne, M.J., Dacke, M., and Baird, E. (2016). A snapshot-based mechanism for celestial orientation. *Curr. Biol.* 26, 1456–1462.
13. Coyne, J.A., Boussy, I.A., Prout, T., Bryant, S.H., Jones, J.S., and Moore, J.A. (1982). Long-distance migration of *Drosophila*. *Am. Nat.* 119, 589–595.
14. Leitch, K.J., Ponce, F.V., Dickson, W.B., van Breugel, F., and Dickinson, M.H. (2021). The long-distance flight behavior of *Drosophila* supports an agent-based model for wind-assisted dispersal in insects. *Proc. Natl. Acad. Sci. USA* 118, e2013342118.
15. Barrows, W.M. (1907). The reactions of the Pomace fly, *Drosophila ampeleophila* Loew, to odorous substances. *J. Exp. Zool.* 4, 515–537.
16. Budick, S.A., and Dickinson, M.H. (2006). Free-flight responses of *Drosophila melanogaster* to attractive odors. *J. Exp. Biol.* 209, 3001–3017.
17. van Breugel, F., and Dickinson, M.H. (2014). Plume-tracking behavior of flying *Drosophila* emerges from a set of distinct sensory-motor reflexes. *Curr. Biol.* 24, 274–286.
18. Heinrich, R. (2002). Impact of descending brain neurons on the control of stridulation, walking, and flight in Orthoptera. *Microsc. Res. Tech.* 56, 292–301.
19. Cheong, H.S.J., Eichler, K., Stüerner, T., Asinof, S.K., Champion, A.S., Marin, E.C., Oram, T.B., Sumathipala, M., Venkatasubramanian, L., Namiki, S., et al. (2023). Transforming descending input into behavior: the organization of premotor circuits in the *Drosophila* male adult nerve cord connectome. *bioRxiv*.
20. Hsu, C.T., and Bhandawat, V. (2016). Organization of descending neurons in *Drosophila melanogaster*. *Sci. Rep.* 6, 20259.
21. Namiki, S., Dickinson, M.H., Wong, A.M., Korff, W., and Card, G.M. (2018). The functional organization of descending sensory-motor pathways in *Drosophila*. *eLife* 7, e34272.
22. Kohatsu, S., Koganezawa, M., and Yamamoto, D. (2011). Female contact activates male-specific interneurons that trigger stereotypic courtship behavior in *Drosophila*. *Neuron* 69, 498–508.
23. von Philipsborn, A.C., Liu, T., Yu, J.Y., Masser, C., Bidaye, S.S., and Dickinson, B.J. (2011). Neuronal control of *Drosophila* courtship song. *Neuron* 69, 509–522.
24. Bidaye, S.S., Machacek, C., Wu, Y., and Dickinson, B.J. (2014). Neuronal control of *Drosophila* walking direction. *Science* 344, 97–101.
25. Rayshubskiy, A., Holtz, S.L., D'Alessandro, I., Li, A.A., Vanderbeck, Q.X., Haber, I.S., Gibb, P.W., and Wilson, R.I. (2020). Neural circuit mechanisms for steering control in walking *Drosophila*. *bioRxiv*.
26. von Reyn, C.R., Breads, P., Peek, M.Y., Zheng, G.Z., Williamson, W.R., Yee, A.L., Leonardo, A., and Card, G.M. (2014). A spike-timing mechanism for action selection. *Nat. Neurosci.* 17, 962–970.
27. Wyman, R.J., Thomas, J.B., Salkoff, L., and King, D.G. (1984). The *Drosophila* giant fiber system. In *Neural Mechanisms of Startle Behavior*, R.C. Eaton, ed. (Springer), pp. 133–161.

28. Ache, J.M., Namiki, S., Lee, A., Branson, K., and Card, G.M. (2019). State-dependent decoupling of sensory and motor circuits underlies behavioral flexibility in *Drosophila*. *Nat. Neurosci.* 22, 1132–1139.
29. Schnell, B., Ros, I.G., and Dickinson, M.H. (2017). A descending neuron correlated with the rapid steering maneuvers of flying *Drosophila*. *Curr. Biol.* 27, 1200–1205.
30. Ferris, B.D., Green, J., and Maimon, G. (2018). Abolishment of spontaneous flight turns in visually responsive *Drosophila*. *Curr. Biol.* 28, 170–180.e5.
31. Dorkenwald, S., Matsliah, A., Sterling, A.R., Schlegel, P., Yu, S., McKellar, C.E., Lin, A., Costa, M., Eichler, K., Yin, Y., et al. (2023). Neuronal wiring diagram of an adult brain. *bioRxiv*.
32. Schlegel, P., Yin, Y., Bates, A.S., Dorkenwald, S., Eichler, K., Brooks, P., Han, D.S., Gkantia, M., dos Santos, M., Munnely, E.J., et al. (2023). A consensus cell type atlas from multiple connectomes reveals principles of circuit stereotypy and variation. *bioRxiv*.
33. Scheffer, L.K., Xu, C.S., Januszewski, M., Lu, Z., Takemura, S.Y., Hayworth, K.J., Huang, G.B., Shinomiya, K., Maitlin-Shepard, J., Berg, S., et al. (2020). A connectome and analysis of the adult *Drosophila* central brain. *eLife* 9, e57443.
34. Dorkenwald, S., McKellar, C.E., Macrina, T., Kernitz, N., Lee, K., Lu, R., Wu, J., Popovich, S., Mitchell, E., Nehoran, B., et al. (2022). FlyWire: online community for whole-brain connectomics. *Nat. Methods* 19, 119–128.
35. Luan, H., Diao, F., Scott, R.L., and White, B.H. (2020). The *Drosophila* split Gal4 system for neural circuit mapping. *Front. Neural Circuits* 14, 603397.
36. Censi, A., Straw, A.D., Sayaman, R.W., Murray, R.M., and Dickinson, M.H. (2013). Discriminating external and internal causes for heading changes in freely flying *Drosophila*. *PLoS Comput. Biol.* 9, e1002891.
37. Muijres, F.T., Elzinga, M.J., Melis, J.M., and Dickinson, M.H. (2014). Flies evade looming targets by executing rapid visually directed banked turns. *Science* 344, 172–177.
38. Olberg, R.M. (1983). Pheromone-triggered flip-flopping interneurons in the ventral nerve cord of the silkworm moth, *Bombyx mori*. *J. Comp. Physiol.* 152, 297–307.
39. Homberg, U. (1994). Flight-correlated activity changes in neurons of the lateral accessory lobes in the brain of the locust *Schistocerca gregaria*. *J. Comp. Physiol. A* 175, 597–610.
40. Steinbeck, F., Adden, A., and Graham, P. (2020). Connecting brain to behaviour: a role for general purpose steering circuits in insect orientation? *J. Exp. Biol.* 223, jeb212332.
41. Kim, H., Park, H., Lee, J., and Kim, A.J. (2023). A visuomotor circuit for evasive flight turns in *Drosophila*. *Curr. Biol.* 33, 321–335.e6.
42. Lima, S.Q., and Miesenböck, G. (2005). Remote control of behavior through genetically targeted photostimulation of neurons. *Cell* 121, 141–152.
43. Isaacman-Beck, J., Paik, K.C., Wienecke, C.F.R., Yang, H.H., Fisher, Y.E., Wang, I.E., Ishida, I.G., Maimon, G., Wilson, R.I., and Clandinin, T.R. (2020). SPARC enables genetic manipulation of precise proportions of cells. *Nat. Neurosci.* 23, 1168–1175.
44. Govorunova, E.G., Sineshchekov, O.A., Janz, R., Liu, X., and Spudich, J.L. (2015). Natural light-gated anion channels: a family of microbial rhodopsins for advanced optogenetics. *Science* 349, 647–650.
45. Goyal, L., McCall, K., Agapite, J., Hartwig, E., and Steller, H. (2000). Induction of apoptosis by *Drosophila* reaper, hid and grim through inhibition of IAP function. *EMBO J.* 19, 589–597.
46. Bender, J.A., and Dickinson, M.H. (2006). Visual stimulation of saccades in magnetically tethered *Drosophila*. *J. Exp. Biol.* 209, 3170–3182.
47. Eckstein, N., Bates, A.S., Champion, A., Du, M., Yin, Y., Schlegel, P., Lu, A.K.Y., Rymer, T., Finley-May, S., Paterson, T., et al. (2020). Neurotransmitter classification from electron microscopy images at synaptic sites in *Drosophila melanogaster*. *bioRxiv*.
48. Lesser, E., Azevedo, A.W., Phelps, J.S., Elabbady, L., Cook, A., Mark, B., Kuroda, S., Sustar, A., Moussa, A., Dallmann, C.J., et al. (2023). Synaptic architecture of leg and wing motor control networks in *Drosophila*. *bioRxiv*.
49. Hulse, B.K., Haberkern, H., Franconville, R., Turner-Evans, D., Takemura, S.Y., Wolff, T., Noorman, M., Dreher, M., Dan, C., Parekh, R., et al. (2021). A connectome of the *Drosophila* central complex reveals network motifs suitable for flexible navigation and context-dependent action selection. *eLife* 10, e66039.
50. Korn, H., and Faber, D.S. (2005). The Mauthner cell half a century later: a neurobiological model for decision-making? *Neuron* 47, 13–28.
51. Strausfeld, N.J., and Hirth, F. (2013). Deep homology of arthropod central complex and vertebrate basal ganglia. *Science* 340, 157–161.
52. Kanzaki, R. (1998). Coordination of wing motion and walking suggests common control of zigzag motor program in a male silkworm moth. *J. Comp. Physiol. A* 182, 267–276.
53. Reiser, M.B., and Dickinson, M.H. (2008). A modular display system for insect behavioral neuroscience. *J. Neurosci. Methods* 167, 127–139.
54. Dionne, H., Hibbard, K.L., Cavallaro, A., Kao, J.C., and Rubin, G.M. (2018). Genetic reagents for making split-GAL4 lines in *Drosophila*. *Genetics* 209, 31–35.
55. Pfeiffer, B.D., Ngo, T.T., Hibbard, K.L., Murphy, C., Jenett, A., Truman, J.W., and Rubin, G.M. (2010). Refinement of tools for targeted gene expression in *Drosophila*. *Genetics* 186, 735–755.
56. Tirian, L., and Dickson, B.J. (2017). The VT GAL4, LexA, and split-GAL4 driver line collections for targeted expression in the *Drosophila* nervous system. *bioRxiv*.
57. Clements, J., Goia, C., Hubbard, P.M., Kawase, T., Olbris, D.J., Otsuna, H., Svirskas, R., and Rokicki, K. (2022). NeuronBridge: an intuitive web application for neuronal morphology search across large data sets. *bioRxiv*.
58. Nern, A., Pfeiffer, B.D., and Rubin, G.M. (2015). Optimized tools for multi-color stochastic labeling reveal diverse stereotyped cell arrangements in the fly visual system. *Proc. Natl. Acad. Sci. USA* 112, E2967–E2976.
59. Cook, K.R., Parks, A.L., Jacobus, L.M., Kaufman, T.C., and Matthews, K.A. (2010). New research resources at the Bloomington *Drosophila* Stock Center. *Fly* 4, 88–91.
60. Namiki, S., Ros, I.G., Morrow, C., Rowell, W.J., Card, G.M., Korff, W., and Dickinson, M.H. (2022). A population of descending neurons that regulates the flight motor of *Drosophila*. *Curr. Biol.* 32, 1189–1196.e6.
61. Guizar-Sicairos, M., Thurman, S.T., and Fienup, J.R. (2008). Efficient sub-pixel image registration algorithms. *Opt. Lett.* 33, 156–158.
62. Suver, M.P., Huda, A., Iwasaki, N., Safarik, S., and Dickinson, M.H. (2016). An array of descending visual interneurons encoding self-motion in *Drosophila*. *J. Neurosci.* 36, 11768–11780.
63. Kim, A.J., Fitzgerald, J.K., and Maimon, G. (2015). Cellular evidence for efference copy in *Drosophila* visuomotor processing. *Nat. Neurosci.* 18, 1247–1255.
64. Boggs, P.T., Spiegelman, C.H., Donaldson, J.R., and Schnabel, R.B. (1988). A computational examination of orthogonal distance regression. *J. Econ.* 38, 169–201.
65. Zheng, Z., Lauritzen, J.S., Perlman, E., Robinson, C.G., Nichols, M., Milkie, D., Torrens, O., Price, J., Fisher, C.B., Sharifi, N., et al. (2018). A complete electron microscopy volume of the brain of adult *Drosophila melanogaster*. *Cell* 174, 730–743.e22.
66. Buhmann, J., Sheridan, A., Malin-Mayor, C., Schlegel, P., Gerhard, S., Kazimiers, T., Krause, R., Nguyen, T.M., Heinrich, L., Lee, W.-C.A., et al. (2021). Automatic detection of synaptic partners in a whole-brain *Drosophila* electron microscopy data set. *Nat. Methods* 18, 771–774.
67. Heinrich, L., Funke, J., Pape, C., Nunez-Iglesias, J., and Saalfeld, S. (2018). Synaptic cleft segmentation in non-isotropic volume electron microscopy of the complete *Drosophila* brain. *Lecture Notes in Computer Science. In Medical Image Computing and Computer Assisted Intervention – MICCAI, A.F. Frangi, J.A. Schnabel, C. Davatzikos, C. Alberola-López, and G. Fichtinger, eds. (Springer International Publishing), pp. 317–325.*

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse mAb anti-Bruchpilot (nc82)	Developmental Studies Hybridoma Bank	nc82; RRID: AB_2314866
Rabbit polyclonal anti-DsRED	ThermoFisher Scientific	Cat #: PA1-986; RRID: AB_560939
Chicken anti-GFP	ThermoFisher Scientific	Cat #: A10262; RRID: AB_2534023
Cy5 goat anti-mouse	Jackson ImmunoResearch	Cat #: 115-175-166; RRID: AB_2338714
Alexa Fluor 488 goat anti-chicken	ThermoFisher Scientific	Cat #: A32931; RRID: AB_2762843
Alexa Fluor 546 goat anti-rabbit	ThermoFisher Scientific	Cat #: A-11035; RRID: AB_2534093
Chemicals, peptides, and recombinant proteins		
All-trans-retinal	Sigma-Aldrich	CAS: 116-31-4
Triton X-100	Sigma-Aldrich	X100
32% Paraformaldehyde	Electron Microscopy Sciences	15714-S
Vectashield Plus	Vectorlabs	H-1900
Deposited data		
Raw and analyzed data	This paper	Mendeley Data: https://doi.org/10.17632/wx3v67wb3w.1
Experimental models: Organisms/strains		
<i>D. melanogaster</i> : w[*]; P{20XUAS-IVS-CsChrimson.mVenus}attP2	Bloomington Drosophila Stock Center	RRID: BDSC_55136
<i>D. melanogaster</i> : w[1118]; 20XUAS-GtACR1-EYFPVK00005	Gift from A. Mauss	N/A
<i>D. melanogaster</i> : w[1118]; P{20XUAS-IVS-jGCaMP7f}su(Hw)attP5;	Bloomington Drosophila Stock Center	RRID: BDSC_80906
<i>D. melanogaster</i> ::; P{w[+mC]=UAS-tdTom.S}3	Gift from D. Anderson	N/A
<i>D. melanogaster</i> : UAS-OpGCaMP7f; UAS-tdTomato	Constructed from above two lines	N/A
<i>D. melanogaster</i> : w[1118]; P{y[+t7.7]w[+mC]=VT025718-p65.AD}attP40/CyO; +	Bloomington Drosophila Stock Center	RRID: BDSC_72300
<i>D. melanogaster</i> : w[1118]; +; P{y[+t7.7]w[+mC]=R56G08-GAL4.DBD}attP2	Bloomington Drosophila Stock Center	RRID: BDSC_69592
<i>D. melanogaster</i> : VT025718.AD; R56G08.DBD	Constructed from above two lines	N/A
<i>D. melanogaster</i> ::20XUAS-SPARC2-I-Syn21-CsChrimson::tdTomato-3.1;	Bloomington Drosophila Stock Center	RRID: BDSC_84144
<i>D. melanogaster</i> ::20XUAS-SPARC2-S-Syn21-CsChrimson::tdTomato-3.1;	Bloomington Drosophila Stock Center	RRID: BDSC_84145
<i>D. melanogaster</i> : P{nSyb-IVS-PhiC31}attP18; S/CyO; Pri/TM6B	Bloomington Drosophila Stock Center	RRID: BDSC_84151
<i>D. melanogaster</i> : nSyb-IVS-PhiC31; VT025718.AD/CyO; R56G08.DBD/TM6B	Constructed from above and DNae014 split driver.	N/A
<i>D. melanogaster</i> : nSyb-IVS-PhiC31; VT013121.AD/CyO; R45H03.DBD/TM6B	Constructed from two above and DNb01 split driver.	N/A
<i>D. melanogaster</i> : SS02383; VT013121.AD; R45H03.DBD	Gift from G. Card, W. Korff, S. Namiki	RRID: BDSC_75888
<i>D. melanogaster</i> : UAS-hid,UAS-reaper;;	Gift from D. Allan	N/A
<i>D. melanogaster</i> : w[1118]; P{y[+t7.7]w[+mC]=R64G09-p65.AD}attP40; MKRS/TM6B, Tb[1]	Bloomington Drosophila Stock Center	RRID: BDSC_71205
<i>D. melanogaster</i> : w[1118]; P{y[+t7.7]w[+mC]=VT044519-GAL4.DBD}attP2	Bloomington Drosophila Stock Center	RRID: BDSC_75123

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
<i>D. melanogaster</i> : R64G09.AD; VT044519.DBD	Constructed from above two lines	N/A
<i>D. melanogaster</i> : w[1118] P[y[+t7.7] w[+mC]=10xUAS-IVS-myr::smGdP-HA]attP18 P[y[+t7.7] w[+mC]=13xLexAop2-IVS-myr::smGdP-V5]su(Hw)attP8; sna[Scd]/CyO; P[y[+t7.7] w[+mC]=R31E10-GAL4.DBD]attP2	Bloomington Drosophila Stock Center	RRID: BDSC_70311
<i>D. melanogaster</i> : R64G09.AD; R31E10.DBD	Constructed from above line and RRID: BDSC_71205	N/A
<i>D. melanogaster</i> : y[1] w[*] P[y[+t7.7] w[+mC]=UAS-myrGFP.QUAS-mtdTomato-3xHA]su(Hw)attP8	Bloomington Drosophila Stock Center	RRID: BDSC_77479
Software and algorithms		
Kinefly wing tracking software	Reiser and Dickinson ⁵³	https://github.com/ssafarik/Kinefly
Magnetotether fly tracking software	W. Dickson	https://github.com/wilddickson/find_fly_angle
Python	https://www.python.org	RRID: SCR_008394
Matplotlib	https://matplotlib.org	RRID: SCR_008624
Fiji	https://fiji.sc/	RRID:SCR_002285
Other		
UV-activated cement	Henkel	Loctite 3972TM
Minutien pins	Fine Science Tools	26002-10
Neodymium ring magnet	Grainger	10E804
Rod magnet	Magcraft	NSN0750/N40
Digital camera	Basler	acA640-120 gm
Digital camera	FLIR	BFS-U3-16S2M-CS
IR light source LED	Thorlabs	M850F2
IR LED ring	W. Dickson	https://github.com/wilddickson/magneto_ir_led_ring
IR light source driver	Thorlabs	LEDD1B
Panel controller	IO Rodeo	Panels display controller unit
Indigo, Skelton Exotic Sangria, and Cyan transmission filters.	Roscolux	Roscolux numbers 59, 39, and 4390.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to Michael H. Dickinson (flyman@caltech.edu).

Materials availability

All new fly lines generated for this paper are listed in the [key resources table](#).

Data and code availability

- All data in main text figures and their associated supplementary figures have been deposited on Mendeley Data at <https://doi.org/10.17632/wx3v67wb3w.1> and will be publicly available upon the date of publication. All new fly lines generated for this study are mentioned in the manuscript text and listed in the key resources table. All original code for automated ROI analysis of 2-photon imaging data has been deposited on Mendeley Data: <https://doi.org/10.17632/wx3v67wb3w.1> and will be publicly available upon the date of publication.
- All original code used for the Kinefly wing tracking and magnetotether body tracking software is publicly available on Github at: <https://github.com/ssafarik/Kinefly> and https://github.com/wilddickson/find_fly_angle, respectively. Any additional information required to reanalyze the data reported in this paper is available upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

All experiments were conducted using 2-to-5-day-old adult female *Drosophila melanogaster*. To specifically target DNae014 cells, we constructed a split-GAL4 line using the hemidriviers VT025718-GAL4.AD to drive the activation domain, and R56G08-GAL4.DBD to drive the DNA-binding domain, resulting in split-GAL4[VT025718.AD; R56G08.DBD]^{21,54–56} (Video S1). Similarly, we created two split-GAL4 drivers that target VES041 cells, split-GAL4[R64G09.AD; VT044519.DBD] and split-GAL4[R64G09.AD; R31E10.DBD]. Candidate hemi-drivers followed from screening databases of standardized MCFO confocal scans of expression patterns of GAL4 drivers.^{57,58} All hemidriviers were obtained from the Bloomington Drosophila Stock Center.⁵⁹ For simplicity, we refer to these specific drivers by their primary targeted neurons, while acknowledging the presence of off-target cells (Videos S1, S2, and S3). We used the split-GAL4 driver SS02383 to target DNb01 cells.²¹ We list off-target cells in the central brain for every driver line in the captions of all supplemental videos. For functional imaging experiments, we crossed males of either the DNae014 split driver or the DNb01 split driver to w⁺;UAS-tdTomato;UAS-GCaMP7f female flies.⁶⁰ For stochastic optogenetic activation experiments, we crossed males from the respective split-GAL4 lines combined with nSyb-PhiC31 (X) with virgin females carrying the intermediate or sparse variants of 20XUAS-SPARC-Syn21-CsChrimson::tdTomato-3.1.⁴³ For VES041 optogenetic activation experiments, we crossed males from the two VES041 split drivers with females carrying 20XUAS-CsChrimson-mVenus on the third chromosome. For temporal silencing experiments, we drove expression of the light-gated GtACR1 channel by crossing males of the DNae014 split driver with females carrying 20XUAS-GtACR1-EYFP on the third chromosome.⁴⁴ For genetic ablation experiments, we drove expression of the pre-apoptotic genes *reaper* and *hid* by using female offspring from crosses between males from each of the DNae014 and DNb01 split drivers with females carrying UAS-rpr, *hid* on the first chromosome.⁴⁵ We reared the progeny of experimental crosses on standard cornmeal fly food, supplemented with additional yeast. For optogenetic activation experiments, larvae were reared in food supplemented with 0.2 mM all trans-Retinal (ATR) (Sigma-Aldrich). For the first two days following eclosion, adult flies were transferred to bottles with standard cornmeal fly food supplemented with 0.4 mM ATR.

METHOD DETAILS

Anatomy

The central nervous system of 3-to-5-day-old female adult progeny were dissected at room temperature (~20°C) in 1X PBS (Sigma) and fixed for 25 min in 4% paraformaldehyde diluted in 1X PBS, and washed 3 times for 15 min in 1X PBS and subsequently 5 times for 15 min in 0.3% PBS-T (1X PBS with 0.3% Triton-X). Samples were then blocked in 7.5% normal goat serum diluted in 0.3% PBS-T for at least 1 hour. The samples were incubated with primary antibodies (mouse nc82 supernatant at 1:30, rabbit polyclonal anti-DsRED at 1:2000 or chicken anti-GFP 1:1000, diluted in blocking solution) for 4 hours at 20°C and 3 days at 4°C, followed by incubation with secondary antibodies (Cy5 goat anti-mouse at 1:300, and Alexa Fluor 546 goat anti-rabbit at 1:1000 or Alexa Fluor 488 goat anti-chicken at 1:1000, diluted in blocking solution) for 4 hours at 20°C and 3 days at 4°C. Samples were imaged using a confocal microscope (Zeiss LSM 880) with a 40X water objective. Optical sections were taken at 1-μm intervals with resolutions of 512X512 or 1024X1024 pixels. The samples were mounted on microscope slides with spacers using Vectashield Plus (VectorLabs).

Functional imaging

Tethered, flying flies were imaged using a galvanometric scan mirror-based 2-photon microscope (Thorlabs) equipped with a Nikon CFI apochromatic, near-infrared objective water-immersion lens (40x mag., 0.8 N.A., 3.5 mm W.D.), as previously described.^{29,60} GCaMP7f and tdTomato fluorescence signals were imaged in the arbors of the left and right DNae014 or DNb01 cells in the SPS neuropil in the brain. Imaging parameters varied depending on the preparation; we either acquired 72 x 36 μm images with 128 x 64 pixel resolution at 13.1 Hz, or in a few cases, 36 x 36 μm images with 96 x 96 pixel resolution at 10.6 Hz (Figure S1). To correct for brain motion in the x-y plane, we registered both channels for each frame using a cross-correlation between each tdTomato image and the trial-averaged image.⁶¹ A field of view (FOV) centered along the bilateral axis was selected based on tdTomato expression. Regions of interest (ROIs) to capture signals of the left and right cells were automatically determined based on GCaMP7f fluorescence pixel variance and covariance with turning behavior, refining signal-to-background ratios and reducing cross talk between contralateral DNb01 cell clusters (see below for details). To account for motion in the z-axis, we normalized GCaMP7f fluorescence to tdTomato fluorescence. To find the baseline fluorescence (F_0), we subtracted the background fluorescence, defined as the 20% dimmest pixels of the entire FOV, from the mean of the lowest 5% fluorescence values in a trial. To standardize neuronal activity, we normalized baseline-subtracted fluorescence, F_t , to the maximum observed for each individual fly in each ROI: $\Delta F/F = (F_t - F_0) / (F_t - F_0)_{\max}$. Neural activity during spontaneous flight turns was recorded in 90 s trials in darkness (i.e. the microscope was enclosed in a light-tight enclosure and the visual arena was turned off). To measure responses to looming, flies were presented with an expanding stimulus created using a 96 x 32 pixel arena that covered 216° of azimuth and ~74° of elevation.⁵³ Looming stimuli simulated a 0.15 m diameter disk approaching at 1.5 ms⁻¹ from a starting position of 1.5 m away from the fly; each trial ended when the stimulus subtended an angle of 60°. Looming stimuli were presented every 6 seconds from one of five directions in pseudo-random order. To reduce light pollution, we shifted the spectral peak of the visual stimuli from 470 nm to 450 nm by covering the arena panels with transmission filters (one sheet of Roscolux no. 59 Indigo, two sheets of no. 39 Skelton Exotic Sangria, and two sheets of no. 4390 Cyan). We tracked left and right wing-stroke amplitudes with a machine vision system, Kinefly, as described before.⁶²

Functional Imaging, regions of interest

We refined the imaging ROIs for functional imaging via an automated process that used the covariance between wingstroke amplitude and pixel variance in GCaMP7f fluorescence, which we confirmed via direct inspection of cell anatomy (Figure S1). Initially, we manually assigned ROIs to capture activity of left and right cells based on visual assessment of tdTomato and GCaMP7f intensities. For the tdTomato signal, we determined the maximum intensity projection in the red channel across all images for each flight trial. For the GCaMP7f signal, we determined the standard deviation for every pixel across each flight trial. We then set a threshold for GCaMP7f intensity within the interrogation areas to identify active pixels and created a binary mask for the left and right cells (Figure S1B). These manually constructed ROIs were used to calculate normalized fluorescence signals ($\Delta F/F$) as described earlier. Relationships between neural activity and behavior were determined using $\Delta F/F$ based on manual ROIs through cross-correlation with wingstroke amplitude (Figure S1C). The $\Delta F/F$ signal in the manually assigned ROI correlated positively with contralateral wingstroke amplitude, and contralateral minus ipsilateral wingstroke amplitude, for both DNae014 and DNb01. To refined single-cell ROIs, we constructed pixel correlograms, based on time-varying fluorescence within a small set of pixels assessed to belong to a single cell, or, for automatic assignment of DNae014 and DNb01 morphology, we cross-correlated the contralateral minus ipsilateral wingstroke amplitude with all pixel fluorescence values from the GCaMP7f channel in a given flight trial. Top ranked pixels from the resulting correlograms were thresholded and binarized to create ROIs for the left and right cells. These ROIs were visually validated against cellular anatomy (Figure S1E). Automatic ROIs were used to obtain single cell GCaMP7f $\Delta F/F$ separately for DNae014 and DNb01 trials, following the same procedures described earlier.

Saccade detection

Fictive saccades were detected using a threshold-based classifier²⁹ operating on the temporal derivative of the bilateral difference in wingstroke amplitude. We used a threshold of twice the measured standard deviation of the derivative for both the functional imaging setup and the rigid tether arena. To avoid tracking errors, we excluded derivative values exceeding 420° s^{-1} and cases in which stroke amplitude fell below 60° . To avoid false positives, caused by rapid changes in wingstroke angles at the end of saccades, we implemented a 0.25 s refractory period following each saccade as in earlier studies.⁶³ Wingstroke amplitude signals were smoothed using a Kalman filter that approximated a 4th-order Butterworth with a low-pass cut-off at 6 Hz. Saccades were classified similarly for magnetotether data, using the time derivative of the body angle and a threshold of 1.5 standard deviations (Figure 4E). Peak saccade speeds in the magnetotether were detected by identifying extrema in body angle velocity after applying a Savitsky-Golay filter ($< 9 \text{ Hz}$). We normalized peak saccade speed to expected peak speed for a given saccade magnitude, based on the measured relationship between saccade speed and saccade magnitude for control flies. Specifically, we normalized all saccade data to the orthogonal-distance regression (ODR)⁶⁴ between peak speed and magnitude for all saccades measured in empty-vector control flies (Figure 4I). We used an 2nd-order polynomial in the ODR because it matched the curvilinear relationship better than a simple line.

Optogenetic activation experiments

Tethered flies were positioned for imaging with a digital camera and backlit with infrared light. Prior to CsChrimson activation, we presented flies with an 8 s visual starfield pattern oscillating sinusoidally about the vertical axis through two full cycles of 216° in amplitude. Wingstroke amplitudes were tracked using Kinefly software.⁶² For optogenetic activation in the rigid tether setup, we guided a fiber optic from a 617 nm LED source (M617F2, Thorlabs) through a microscope objective, adjusted to create a 0.5 mm spot on the posterior side of the head (Figures 3D and S6), with a power density of $\sim 0.2 \text{ mW mm}^{-2}$. For optogenetic activation in the magnetotether, we delivered 623 nm light via a custom-built LED ring (Figure 6B). For unilateral activation experiments of either DNae014 or DNb01 (Figures 3D–3I) we elicited 6 responses to a 0.25 s light pulse at 10 s intervals. Bilateral CsChrimson activation experiments of VES041 involved three 15 s periods of excitation surrounded by 15 s periods of no excitation. If a fly stopped flying, we collected additional trials. For each fly, the three first presentations with continuous flight were included for further analysis.

Unilateral genetic targeting of CsChrimson

To stochastically target the optogenetic effector CsChrimson tagged with tdTomato within the split-GAL4 driver lines (for either DNae014 or DNb01), we employed Sparse Predictive Activity through Recombinase Competition (SPARC).⁴³ We crossed parental lines containing pan-neuronal phi-C31 and the split-GAL4 of interest with the appropriate SPARC line (20XUAS-SPARC2-I-Syn21-CsChrimson::tdTomato-3.1 for intermediate labeling or (20XUAS-SPARC2-S-Syn21-CsChrimson::tdTomato-3.1 for sparse labeling). The resulting flies expressed CsChrimson and a fluorescent marker, tdTomato, in either the left cell, the right cell, both cells, or no cells. Expression patterns were manually scored using fluorescence microscopy after behavioral measurements, and a subset of flies underwent immunostaining to obtain representative confocal microscopy images (Figure 3E). All experiments were performed in a blinded fashion, with expression patterns naïvely scored behavioral data collection. These experiments were performed on intact, rigidly tethered flies.

Connectomics

We identified the DNae014 and DNb01 cells in publicly available connectomes.^{31–33} Comparison to fly brain connectomes indicated at least four DNs in the subclass of DNa neurons²¹ that roughly resembled DNae014, but one neuron matched the morphology of the cell in our split-GAL4 line most closely (G. Jefferis and M. Costa, personal communication). The hemibrain dataset contains two neurons labeled DNb01, with one matching the expression pattern of the cell we refer to a DNb01 based on the description of Namiki and

coworkers,²¹ which is under control of the split-GAL4 driver we used in our experiments, SS02383. We conducted a connectivity analysis based on both the FlyWire connectome based on the FAFB EM dataset, and the hemibrain connectome datasets^{31–34,65} (see <https://flywire.ai/>). Neurons in FlyWire were considered of the same type following labeling in the hemibrain and/or due to similarity in both anatomy and connectivity (see Figure S3A). We identified saccade-generating unit input neuron types (SGUIs) that innervated both unilateral DNae014 and DNb01 cells with combined connection strengths above a threshold where the product of synapse counts onto DNae014 and DNb01 cells exceeded the square of 75. This threshold was objectively determined as the lowest threshold at which the set of SGUIs to the left SGU were identical to the symmetrical set of SGUIs to the right SGU (Figures S3B and S3C). Putative saccade network neurons were labeled according to the FAFB or hemibrain connectomes (Table S3). To assess interconnectivity between identified neurons, we included direct connections that contained over 37 synapses (Figures S3 and 5), as well as indirect connections between left and right SGUIs that contained over 75 synapses. For these inter-neurons connecting left and right SGUIs, we applied a higher threshold to exclude more likely false-positive connections due to high local innervation density. Neurotransmitter types were predicted using a machine-learning synapse classification algorithm based on confidence scores and synapse cleft width.^{47,66,67}

QUANTIFICATION AND STATISTICAL ANALYSIS

All experiments were analyzed with custom software written in Python. Sample sizes refer to the number of individuals tested. Variance across individuals was assessed by either standard deviation of individual means (Figures 2D, 4C, 4F, 4H, 4I, 4K, 6E, S1C, S1F, S2D, and S6E) or boot-strapped, 95% confidence intervals (CI) for the mean of the individual means (Figures 1L, 2B, 2C, 3C, 3G–3I, and 4G) S2. Statistical differences between peak responses in the activity of DNs and loom directions were determined with a 2-way ANOVA with cell type and loom direction as the main effects and individual as a random effect (Figure 2D). Statistical differences between controls and DN groups were assessed using Kruskal-Wallis tests with Tukey's HSD method for post-hoc comparisons. For VES041 activation in the magnetotether, paired samples t-tests, we used a Kruskal-Wallis test followed by post-hoc comparisons with Tukey's HSD method to evaluate differences among the three genetic lines during CsChrimson activation (Figure 6E). Significance levels are indicated as ns (not significant), * ($p < 0.05$), ** ($p < 0.01$), or *** ($p < 0.001$) (Figures 2D, 4C, 4F, 4H, 4I, 4K, and 6E).